SW Engineering CSC648
StudySync.me https://github.com/ssohbn/SFSU-csc648-06-Fall26-team04
section 06, team 04
Maeve (team lead)
Henry (frontend lead)
Shreya (scrum master)
Aleks (backend lead)
Jerald (git specialist)
Simran (Ai lead)
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1) Executive Summary:

Our program is a study group finder, organizationally similar to canvas where a user is registered to classes or interests. On studysync you can sync your studying with other people by creating study groups for your interests, which may include school classes or some topic you would like to study with other people. Being a student, it is clear how important studying and doing classwork with other students is. Being with other people working on similar subjects can be motivating to completing schoolwork, and being able to ask each other for help can be useful to understanding new topics. This is what brought us to creating studysync, an app to find and create study groups with other people sharing your interests.

On studysync, users create a profile with a list of interests that they would like to study. Interests can be anything; they can be a class code like “CSC415,” a topic like “beginner-intermediate spanish,” or simply “casual study.” For students using canvas, they could also import class codes as interests using the canvas calendar link, which we use to obtain the classes that students are actively in.

After a user's profile is populated with interests, they can look for study groups, searching by name and/or filter groups by interests to find a study group that suits them. Our ai will also attempt to match students to a suitable study group, based on their interests

Users can also create study groups, where they first select some of their interests to study, give the group a name, and fill out information pertaining to their study group. This information can include a location like an address or a zoom link depending on if the study sessions are being held in person or online, a meeting time, contact information, and a description of what the study group will be like. Obviously people like to study in different ways, so this description is kept open for flexibility. One user might want to have recurring meetings in a public place, and another user might want to have one-off study sessions to cram right before a big midterm.

The flexibility of studysync helps students quickly create or join study groups in whatever interests and scheduling needs they may have. As students creating this platform, we understand the benefits of studying with others, and the heavy flexibility needed to cater to students with varying needs, situations, and deadlines coming both from school pressure and pressure from outside of it.

2. Personas and User stories:

Persona 1: Group Leader

A student who regularly organizes study sessions or class projects. They want to create a Study Group around one or more shared Interests. In their study groups, they act as a Group Admin/Leader and manage the group's information and group members.

Persona 2: Commuter Student

A student with a tight schedule who may only be on campus during certain times. They want to use Find Groups to locate Study Groups related to their classes and interests and find groups whose meeting schedule and location work for them.

Persona 3: Undeclared Student

A student who is exploring possible majors, classes, or academic subjects. They want to add subjects they are interested in to the Interests on a Profile and want to find Study Groups with Users who are already studying those subjects.

Persona 4: Professor

A professor who wants their students to organize study groups independently. They want students to be able to import their classes as Interests based on information on Canvas and use those Interests to find other students and Study Groups related to the class.

Persona 5: Struggling Student

A student who is struggling with class material, motivation, or unreliable tutoring. They want to add their classes as Interests and use Find Groups to locate other Users studying the same material.

Persona 6: Random Interest Learner

A User who wants to meet other people interested in the same subjects, including interests outside of normal school classes. They want to add these subjects to their Profile as Interests and find Study Groups centered around those Interests.

User Stories

US1 : High Priority
As a Commuter Student, I want to search and filter Study Groups by Interest on the Find Groups page so that I can find groups relevant to what I am studying.

US2 : High Priority
As a Struggling Student, I want to find Study Groups matching my class Interests so that I can study with other Users working on the same material.

US3 : Medium Priority
As an Undeclared Student, I want to find Study Groups related to my academic Interests so that I can meet students taking classes or studying subjects that I may pursue in the future.

US4 : Medium Priority
As an Interest-Based Learner, I want to find Study Groups associated with my non-class Interests so that I can meet Users interested in learning or discussing the same subjects.

US5 : High Priority
As a Group Leader, I want to create a Study Group around one or more of my Interests so that other Users can find and join the group.

US6 : High Priority
As a Group Admin, I want to edit my Study Group's meeting schedule, location, description, etc. so that enrolled Users can see current information about the group.

US7 : Medium Priority
As a Group Admin, I want to review Join Requests and approve Users who request to join a Study Group that requires approval.

US8 : High Priority
As a User, I want to freely join a public Study Group or send a Join Request to a group that requires approval so that I can become enrolled in groups that interest me.

US9 : High Priority
As a User, I want to view my Study Groups through the My Groups page so that I can easily access the groups I am currently participating in.

US10 : High Priority
As a User, I want to edit the Interests on my Profile so that StudySync can represent the classes, topics, and subjects that I want to study.

US11 : Medium Priority
As a Professor, I want students to be able to create their interest based on their Canvas so that they can more easily find other Users and Study Groups associated with their classes.

US12 : Medium Priority
As a Struggling Student, I want to view a Study Group's meeting schedule, location, and description before joining so that I can find a study environment that works for me.

3.Data Definitions

User: describes a user profile of our program

- Has a username and a password **ONLY**
- Location (used to filter groups)
- Bio includes interests
- Study Groups they're enrolled in

Has a set of interests and is enrolled in certain groups

Interest: these are, for example, classes or hobbies that study groups are designed around

- Used for recommending groups or describing the topic a study group is made for
- Used to describe a user

Study group: logical group of users and resources related to an interest

- Meeting schedule
- Link to meeting online via Zoom, Google Meet, or other preferred online platforms
- Mailing list?
- Link to a Discord?
- In person meeting location (WITH NIANTIC ROOM SCAN)

User Roles: Group Admin, enrolled user

- Admins
  can edit study group details
  Can approve joins and invite users

- enrolled users, can see study group information

Group publicity: describes whether the group is free to join or needs an application to be approved

Public: can be joined freely
Invite: needs a join request

Join requests are just a one click button, no form.
Admins of the study group can then from request, view profile and approve users, should be casual

Profile page: shows your user profile, editable by the user, or viewable by others

My groups page: used to get to your different groups and view information about a group

Find groups page: used to find study groups to join, can search names and filter by interests

Canvas Calendar Information: Classes, Assignments

4. initial list of functional requirements

1) **us10,us9**

- Sign up and make profile

We need to be able to make profiles to customize content to different users and enroll people in groups

- Profile
    - Bio editing
    - Interest editing
    - General Location

The profile needs to be editable with a bio and interests, and a general location would be useful to filter in person studying

- Pages
    - Profile Page
        - Shows user profile
    - edit profile page
        - used to edit the profile

The profile needs a page to be shown and edited on

Us1, us2

- Group Search Page
- My groups Page

Groups need to be searchable to enroll in new groups. Users need a way to see what groups they are enrolled in.

Us5, us6

- Study group system
    - Group recommendation
    - Group enrollment
    - Editable group information

We need a group enrollment system that allows users to join and leave groups. We also want to recommend groups to users so that they are able to study their interests. Group information should be editable to tailor it to what the study group is about and so that it can be tailored to what type of studier it caters to.

us7

- Admin and user roles
- Study group privacy

We need study group administration so that people can have control over who is in or out of groups. Privacy is important especially to in person events and places where space is limited or simply not public

Us3, us11

- Canvas class import via calendar

Importing classes from canvas calendar makes it really easy to fill out interests as a student and match with other students in your classes. This is very useful for professors to help students students study with eachother if they want to, and it help students find other students from their class when they need help and want to study with someone.

5. Non functional requirements:

1) Must run well on a low powered school provided laptop
2) Must be intuitive to use as a student coming from canvas
3) Must run on firefox
4) Git commits should be read and checked off by other team members

6. Competitive analysis
Competitors

competitor	focus/competitor features	What ours does better/solves	
canvas	Focused on teachers for students in more strict classroom setting	Students cant create groups with eachother	
sfsu tutoring	Organized by school to give students help on classes Limited tutors	Organized by students and people who want to learn Students who just want to study with eachother may not want a specialized tutor, users on our serveice just want a study group together	
studystream	Virtual studying and online focus rooms Kinda creepy	More focused on in-person, campus-based study groups, connecting students from the same school, courses, and interests. More casual	

yeolpumta	Its in korean	Ours is in english	

Our project is more flexible than other available studying platforms and resources, and aims to cater to a more casual studying experience, that is organized by the studiers themselves. Canvas focuses on teachers and teaching classes, whereas our program focuses on students and studying content. We want to provide a space to organize in person and optionally online study rooms, unlike studystream which is purely virtual, and kind of creepy with how its users all have their cameras on. Sfsu tutoring is very official and can lack in tutors, and many students do not want a tutor, but just want another student to study with. Our goals are to cater to casual studying on interests, unlike the goals of the competitors listed above

7) High-level system requirements Briefly provide itemized list of all main SW components such as frameworks, tools and systems to be used, supported browsers and deployment platform (SW and server) to be used. This list is to be the list of approved tools and systems from M0. Any other external (open source) code/API/tool must be listed. If students pursue a 3D spatial app based on Niantic technology, please include Niantic technology here

9.Checklist:
• Team found a time slot to meet outside of the class
Done

• Scrum Master shares meeting minutes with everyone after each meeting (what have been discussed, what to do, what is the issues).
Done

• Github master chosen
Done

• Everyone sets up their local development environment from the team's git repo.
Done

• Team decided and agreed together on using the listed SW tools and deployment server
Done

• Team ready and able to use the chosen back/front-end frameworks.
o For each technology (front/back-end/DB/cloud/AI such as Niantic technology), team decides who will lead the study of each technology and what will be the specific goal of the study within one month from the M1 Announcement.
On track

• Team lead ensured that all team members read the final M1 and agree/understand it before submission
Done